The Use of an 8-Week Mixed-Intensity Interval Endurance-Training Program Improves the Aerobic Fitness of Female Soccer Players

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Abstract

Clark, JE. The use of an 8-week mixed-intensity interval endurance-training program improves the aerobic fitness of female soccer players. *J Strength Cond Res* 24(7): 1773–1781, 2010—The purpose of this study is to examine improvements in cardiorespiratory fitness ($\dot{V}O_2$) after the use of a mixed-intensity interval endurance-training (MI-ET) program in female soccer players, to validate the MI-ET program as an appropriate training regimen to improve cardiorespiratory fitness ($\dot{V}O_2$) in soccer players. 32 female soccer players (average 18.66 $\pm$ 0.31 years) were recruited from a group of currently conditioning local U-19 and college soccer teams and randomly assigned to participate in an 8-week periodized training program that involved either the MI-ET program or the continuation of a current endurance-training (ET) program. Analysis of variance indicates no differences in $\dot{V}O_2$ values within the group of athletes before participating in the exercise program. After the 8 weeks of training, the MI-ET group of athletes had significantly greater average $\dot{V}O_2$ values (62.13 $\pm$ 0.96 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$ vs. 57.27 $\pm$ 1.59 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$), $p = 0.015$, along with a greater group average of change in $\dot{V}O_2$ (12.44 $\pm$ 0.92 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$ vs. 7.72 $\pm$ 0.99 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$), $p < 0.001$. The MI-ET program is shown to be a valid means to improve aerobic fitness as indicated by the MI-ET group exhibiting significantly greater $\dot{V}O_2$ measures after training.

Key Words aerobic training,, intermittent endurance exercise, improvement in $\dot{V}O_2$

Introduction

The general purpose for conditioning protocols is to allow athletes to increase their ability to elicit a maximal effort of performance during competition, for example, athletic prowess, after the employment of the conditioning regimen. To ensure a proper conditioning regimen, the physiological demands of the sport should direct and determine the type of conditioning that the athlete will use to maximize their ability. In the case of soccer, this means the ability to produce a large and sustained workload for a prolonged period of exertion (1,5,9) and has been matched in conditioning programs by the incorporation of endurance-training (ET) and aerobic conditioning exercises (5,7). Traditionally, ET patterns are described as activities, for example, walking, running, or jogging, at a set heart rate ([HR] heart beats per minute [$b \cdot min^{-1}$]) for an elongated duration employed multiple times each week. Recently, there has been a suggestion that the use of maximal effort sprint interval training elicits improvements in aerobic fitness and capacity similar, increases the $\dot{V}O_2$ (ml $O_2 \cdot kg^{-1} \cdot min^{-1}$), to those seen with the traditional ET (2,4,6,10).

When combining the favorable results of sprint-interval training and the analysis of soccer movement, an increased employment of sprint-interval training has become an integrated component of most soccer conditioning programs. Furthering the idea of need for high-intensity training in soccer conditioning, Hoff (7) has reported that within a match play of a game, an athlete's movement and subsequent HR response occasionally reach maximal levels of exertion, which is immediately followed by active recovery within play before the next maximal exertion. However, when analyzed with the pattern of effort responses throughout the game, the commonly used sprint-intensity interval exercises, where a maximal sprint is immediately followed by prolonged inactive period of recovery, does not match the demands seen within a competitive game of soccer.

Hence, a problem arises that even though the sprint-interval ET may elicit an increase in $\dot{V}O_2$ for the soccer athlete, it does not mimic a pattern of movement that will elicit training adaptations necessary for carry over to,

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improvements in functionality of, soccer playing. Thus, the purpose of this study is to examine, and ultimately validate, an interval-training ET program that provides a close mimic to the exertion patterns seen within the playing pattern of movement and exertion of a soccer game. The interval-training program, mixed-intensity interval ET (MI-ET), was developed that would encourage the positive cardiorespiratory responses of both the sprint-interval ET and continuous-moderate ET, while also mimicking the demands of play within the soccer game.

To develop the desired pattern of exercise for conditioning, multiple soccer matches were logged to track the movement patterns, intensities, and time for exertion of players, from various positions, throughout the length of a game. From these logs, it was determined that a pattern of short duration high-intensity aerobic exercise followed immediately by short duration moderate-to-low-intensity aerobic exercises would best mimic the observed movement and maximal cardiorespiratory demands of a soccer match. This observation of the movement patterns accompanied by previous studies (1,7,9), and thus the physiological demand, of the soccer match lead to a supposition that the movement pattern more closely resembles the intervals of exertion and level of intensity seen with training for a long-and middle-distance track runner (2). This pattern was anecdotally tracked via subjective ratings by 10 female soccer players from 2 local senior high schools over a course of 1 competitive season. After repeated trials and the analysis of subjective feedback after each trial, the most convenient and highly reproducible pattern of high-to-low intensities was in 30-second increments.

There are 2 reasons for the selection of the 30-second increments as the baseline for the interval times. First, a training demand was needed that although mimicking the exertion pattern within soccer was more energetically taxing on the athlete than traditional drilling, practicing, or competing, thus allowing for increased cardiorespiratory demand and subsequent adaptations to training (2,8–10). Second, the time increment of 30 seconds approximated the time needed to get from touchline to touchline (as if making a breakout run) and back to a playing position for the players that tested the develop of the time intervals for the program. From these trials, and using the 30-second increments, a program was developed that would incorporate 3 separate patterns of the 30 seconds of varying intensities followed by an equal amount of time within a pattern of active recovery by the athletes. The pattern of exercise was eventually broken into 3 separate time constraints, ranging from 30 seconds of maximal intensity with 30 seconds of active rest, to 60 seconds of near-maximal intensity with 60 seconds of active recovery, and 90 seconds of submaximal intensity with 90 seconds of active recovery.

Even with the support for the incorporation of sprint-interval training into training and conditioning sessions for soccer players (2,4,6,7,10), there exists a possible disjunction between the parameters of sprint-interval and high-intensity interval training to athletic competition in soccer. Therefore, this has led to a necessity to examine an interval-training program that is a closer mimic to the demands undertaken within a competitive soccer event and will therefore provide better carry-over of conditioning to the demands of a competitive soccer game. From this necessity, an interval-training protocol was developed that was added to a concurrent off-season training program for college aged female soccer players, and thus, it is the aim of this study to examine the cardiorespiratory adaptations obtained after the 8-week use of the MI-ET program. This aim was established in an effort to validate its use as an appropriate training modality for soccer players to use to improve aerobic fitness using a parameter that meets and mimics the demands contained within their sport.

Methods

Experimental Approach to the Problem

This examination was established to test the hypothesis that there will be no difference between cardiorespiratory responses ($\dot{V}O_2$, ml $O_2 \cdot kg^{-1} \cdot min^{-1}$) after an 8-week training session for athletes that participate in the novel MI-ET program vs. those athletes that continue to participate in the concurrent ET program within an off-season conditioning program. Further, these cardiovascular responses will compare favorably to the previously reported findings involving the use of sprint-interval ET for cardiorespiratory responses. This analysis will thus provide the evidence and support the fact that the MI-ET program is a valid method for improving cardiorespiratory fitness needed for soccer players after training.

Subjects

Thirty-two female athletes, ranging in age from 16 to 23 years, average age of 18.66 ± 0.31 years, were recruited from the local population of College and Adult Recreational Soccer Clubs (U-19), (Table 1). All volunteers were recruited from a currently training population of athletes in the off-season of their respective soccer programs. The study was conducted per the rules for the college; the group of athletes read and completed informed consent forms, with parental consent when required, for the analysis of changes with addition of the MI-ET program to the off-season conditioning program. After the consent to perform in the study, volunteers were given medical clearance for participation from the team sports medical staff and then finally underwent initial screens for biometrics (height and body weight) before being enrolled into the study. After enrollment into the study, volunteers were randomly assigned to 1 of the 2 training methods, MI-ET or ET. Of the 32 volunteers, 17 (average age of 18.7 ± 0.37 years) were enrolled into a program of MI-ET, whereas the remaining 15 (average age of 18.6 ± 0.51 years) were enrolled to continue to train using the continuous-moderate intensity ET.

The volunteers in the MI-ET group underwent the supervised exercise sessions during a scheduled afternoon time

TABLE 1. The summary of demographic information, age (years), weight (kg), and method of training (ET or MI-ET) for the participants in the 8-week endurance-training program.*

ID #	Age (y)	Weight (kg)	Exercise group a	ID #	Age (y)	Weight (kg)	Exercise group a
1	17	42	MI-ET	17	18	57.6	MI-ET
2	18	46.4	MI-ET	18	18	57.2	MI-ET
3	17	52	MI-ET	19	18	52.8	MI-ET
4	17	50	ET	20	18	57.6	ET
5	19	47.6	ET	21	19	51.6	ET
6	17	52.8	ET	22	18	55.6	MI-ET
7	19	56	MI-ET	23	20	54	ET
8	17	54	ET	24	21	53.6	MI-ET
9	17	52.8	MI-ET	25	22	53.6	MI-ET
10	19	53.2	MI-ET	26	20	60	MI-ET
11	16	49.2	ET	27	21	53.6	ET
12	17	46.8	ET	28	19	53.2	MI-ET
13	17	50	ET	29	20	52.8	ET
14	17	53.6	ET	30	22	54	ET
15	19	56	MI-ET	31	21	54.4	ET
16	17	58	MI-ET	32	22	50.8	MI-ET
Whole group athlete averages							
Age of athlete (y)	18.66 ± 0.31						
Weight of athlete (kg)	52.9 ± 0.66						

*a = exercise group placement; MI-ET = the mixed-intensity interval group; ET = the continuous-moderate endurance group.

where the soccer field was available for training, whereas those in the ET group ran at self-selected times at HR intensities and total time of exercise that corresponded to the previously established concurrent ET program. To monitor the training of the volunteers for both patterns of exercise HR response, b·min^{-1} was continuously monitored via Polar F4 (Polar Electro, Lake Success, NY, USA) that was calibrated for each athlete's individual biometrics. Additionally, all volunteers were advised to continue with the additional exercises within the off-season conditioning program and to make no changes to their diet during the 8 weeks of investigation of the MI-ET program.

Experimental Design

Before, and after the 8-weeks of, the additional training with either the MI-ET or the continuous-moderate intensity ET, players underwent aerobic fitness testing under the guidance and supervision of coaches and researcher via a 12-minute run test of aerobic fitness (3). The 12-minute run test was used to allow coaches with a continuous test–retest procedure to provide for a continuous of monitoring of cardiorespiratory fitness for all athletes within their program.

All testing was performed in groups of 8 volunteers at a time on a 400-m track covered with synthetic surface, where cones were placed at 20-m intervals along the inside border to track to measure increments of the 400 m covered by the athletes at the end of 12-minute running test. Two coaches and the researcher would tally the number of 400-m laps completed by each runner to ensure the accuracy of distance covered. During the last minute of the 12-minute running time, time countdown was given in a 15-second interval and then with a final 10-second countdown at the individual second from 10 to 0 seconds. After completion of the 12-minute running test, the total distance covered by each athlete was calculated based on the number of 400-m laps completed and then the percentage of lap completed (based on the number of 20-m increments completed within that lap). This distance was then converted into kilometers (km) run during the 12 minutes and then inputted into the following equation: $\dot{V}O_2max = (22.351 \times \text{km run}) - 11.288$, that would determine the estimated $\dot{V}O_2max$ for the individual athlete.

Mixed-intensity Interval Training

The MI-ET program used intervals that were organized into a 6-minute cycle of intervals. This pattern was completed as follows: 30-second submaximal jogging followed by 30 seconds of 90–100% of maximal effort ($\dot{V}O_2$), followed by 60 second of submaximal jogging followed by 60 seconds of 80–90% of maximal effort, and finally 90 seconds of submaximal jogging followed by 90 seconds of 70–80% of maximal effort. Within each session, the interval patterns were periodized to be performed over 12 minutes of interval running (2 complete cycles of the interval program) and increased by additional cycles of intervals for the following weeks, increasing to a total time of 36 minutes (6 complete cycles) by the seventh week of training (Table 2). Each session of exercise was repeated 3 times per week with 24–48 hours of recovery between sessions. To provide the appropriate feedback to the

TABLE 2. The description of exercise design and progression for each of the 2 training programs completed over the length of the 8-week periodized ET program.*

	MI-ET group							
Weekly schedule	Monday Intervals	Tuesday Active recovery	Wednesday Intervals	Thursday Active recovery	Friday Intervals	Saturday Active recovery	Sunday Active recovery	
Weekly periodization of time of training and number of repetitions of intervals								
Week	1	2	3	4	5	6	7	8
Time for training (min)	12	18	18	24	30	30	36	36
Total repetitions of intervals	2	3	3	4	5	5	6	6
Intervals for training based on intensity (% $\dot{V}O_2max$) and time								
Interval #	1	2	3	4	5	6	Repeat intervals in order #1–#6 for total time of training per session	
% $\dot{V}O_2max$	50–60	90–100	50–60	80–90	50–60	70–80		
Time (s)	30	30	60	60	90	90		
	Concurrent ET group							
Weekly Schedule	ET	Active recovery	ET	Active recovery	ET	Active recovery	Active recovery	
	Weekly periodization of time of training							
Week	1	2	3	4	5	6	7	8
Time for training (min)	40	40	45	45	45	50	60	60

*MI-ET = mixed-interval endurance training; ET = endurance training

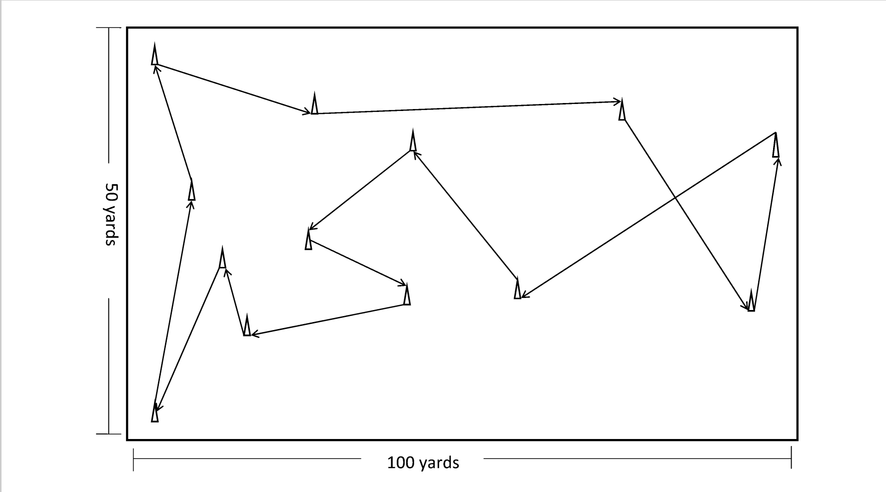

Figure 1. Generalized field layout for cones and changes of directions used during the mixed-intensity interval endurance-training program.

volunteer during each training session, HR response was tracked continuously on the Polar F4 (Polar Electro) monitor that each volunteer wore throughout the exercise session.

All exercise sessions took place on an open grass soccer field that was similar in dimension to a regulation soccer field, with a random-distance layout of cones for volunteers to use as a guide for where to change course of movement throughout the training session (Figure 1). All sessions began with a 5-minute active warm-up that included low-intensity aerobic activity followed by self-selected active stretching activities for each of the volunteers. From the end of the

Table 3. The summary of the pre and posttest aerobic fitness (estimated $\dot{V}O_2$max), along with the absolute and relative percent change of $\dot{V}O_2$ after training, for all participants involved in the 8-week ET program.*

MI-ET group			ET group		
Volunteer ID #	$\dot{V}O_2$ pretest ($ml \cdot kg^{-1} \cdot min^{-1}$)	$\dot{V}O_2$ posttest ($ml \cdot kg^{-1} \cdot min^{-1}$)	Volunteer ID #	$\dot{V}O_2$ pretest ($ml \cdot kg^{-1} \cdot min^{-1}$)	$\dot{V}O_2$ posttest ($ml \cdot kg^{-1} \cdot min^{-1}$)
1	48.78	66.05	4	42.67	48.78
2	48.78	66.05	5	55.26	60.65
3	53.46	66.05	6	57.05	60.65
7	42.67	57.05	8	57.05	60.65
9	48.78	62.45	11	51.66	62.45
10	48.78	60.65	12	49.86	55.26
15	48.78	62.45	13	42.67	48.78
16	51.66	69.64	14	36.55	44.46
17	51.66	66.05	20	36.55	51.66
18	52.02	62.45	21	48.78	58.85
19	51.66	64.25	23	51.66	62.45
22	48.78	60.65	27	55.97	60.65
24	51.66	62.45	29	55.97	58.85
25	53.46	58.85	30	57.05	66.05
26	46.26	58.85	32	44.46	58.85
28	48.78	55.26			
31	48.78	57.05			
Exercise group averages		$\dot{V}O_2$ pretest ($ml \cdot kg^{-1} \cdot min^{-1}$)	$\dot{V}O_2$ posttest ($ml \cdot kg^{-1} \cdot min^{-1}$)	Absolute change in $\dot{V}O_2$ ($ml \cdot kg^{-1} \cdot min^{-1}$)	Relative % of change in $\dot{V}O_2$
ET		49.54 ± 1.89	57.27 ± 1.59	7.72 ± 0.99	0.166 ± 0.03
MI-ET		49.69 ± 1.15†	62.13 ± 0.96‡§	12.44 ± 0.92‡§	0.252 ± 0.03‡§

*MI-ET = mixed-intensity interval endurance-training; ET = endurance training.
†MI-ET group shows no significant difference from the ET group.
‡MI-ET group is on average significantly greater than the ET group.
§MI-ET group based on individual change is significantly greater than the ET group after the 8-week training program.

active warm-up, volunteers would begin to continuously sprint, run, jog, or walk throughout the course of a training session based on the indication for intensity within a given interval. To inform the volunteer of the level of intensity for the interval, a coach would blow a whistle at the beginning and the end of each of the time intervals of variable exertion (1–whistle blow for submaximal recovery; 2–whistle blows for 90–100% maximal; 3–whistle blows for 80–90% maximal; and 4–whistle blows for 70–80% maximal). After each training session, athletes would participate in a 5-minute cool-down including very-low-intensity aerobic activity and self-selected passive, and active, stretching exercises.

Continuous-Moderate Endurance Training

Volunteers that were not assigned to the MI-ET program continued to use the previously prescribed ET program that was being used by the athletes under the guidance of the coaching staff and trainers. This program continued the running program that involved the use of 40–60 minutes of continuous running at a perceived rate of exertion labeled as "moderate-hard" to "hard," 3-times per week (6–8). The only adaptation made to this program during the 8-week study was the use of HR monitoring to judge level of intensity during training and a periodization of time of running to be performed for each training session. To ensure a consistent level of effort throughout the run, each athlete was instructed to maintain a running pace that would force an average an HR response of 60–80% of her individual $\dot{V}O_2max$ (6,8), as monitored via a Polar F4 (Polar Electro) HR monitor, throughout total time of running.

Just as with the MI-ET group, the continuous-moderate ET group would begin each training session with a 5-minute active warm-up that included low-intensity aerobic activity and self-selected active stretching exercises. After the active warm-up, the athletes would begin their running session for that day for the total time allotted at the appropriate level of intensity based on the running schedule. After each running session, athletes would participate in a 5-minute cool-down including very-low-intensity aerobic activity and self-selected passive, and active, stretching exercises. Additionally, the total time of running was periodized as to match the periodization seen with the MI-ET program, where total running time ranged from 40 minutes during the first week to 60 minutes by the eighth week of training (Table 2).

Statistical Analyses

Data were collected after each testing session and inputted into Microsoft Excel™ Spreadsheet for volunteers based on their respective numbers. These data were then analyzed for differences in $\dot{V}O_2$ measures between the groups along with the difference within each of the 2 exercise group differences via a repeated measures analysis of variance (ANOVA) ($p \leq 0.05$) for possible differences between athletes seen before and after the 8-week training session of MI-ET or continuation of the concurrent ET program. Additionally the averages among the 2 groups were analyzed via a paired T-test for mean ($p \leq 0.05$) for possible differences between the group averages for $\dot{V}O_2$ measures after the 8-week training of MI-ET or the continuation of the concurrent ET program.

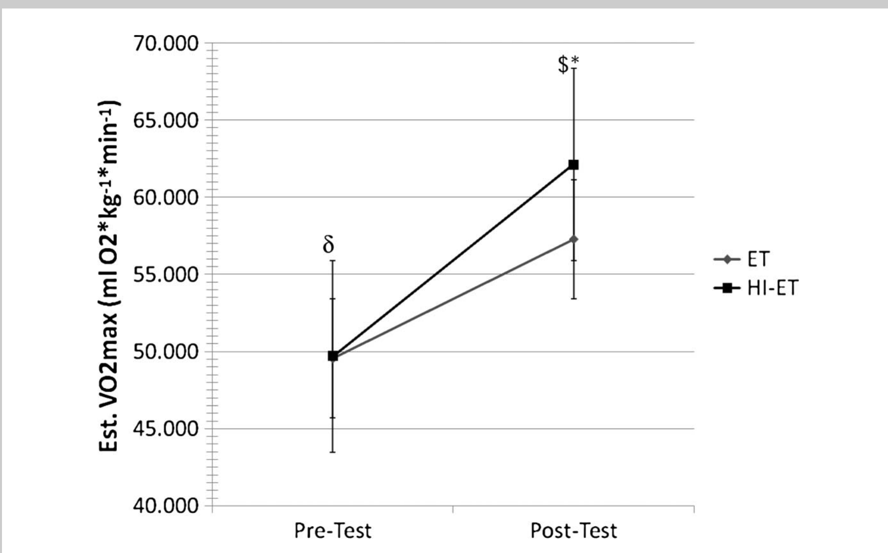

Figure 2. Group average pre and posttest measures of estimated $\dot{V}O_2max$ for the participants of the 8-week endurance training (ET) program broken into the 2 training methods, ET = continuous-moderate, MI-ET = mixed-intensity interval. *Indicates that the MI-ET group is on average significantly greater than the ET group, and $ indicates that the MI-ET group based on individual change is significantly greater than the ET group after the 8-week training program. The δ indicates that the MI-ET group shows no significant difference from the ET group.

Results

The summary of individual and group pre and posttest measures, and the changes estimated $\dot{V}O_2max$ (ml $O_2 \cdot kg^{-1} \cdot min^{-1}$) can be found in Table 3. Analysis of general biometrics indicates no difference between groups for the age or body mass of individuals within the group or between the 2 exercising groups. Analysis of the results for the entire group of athletes shows that there is limited variance within the whole group for the individual pretest $\dot{V}O_2$ ($p = 0.089$), absolute changes in $\dot{V}O_2$ ($p = 0.462$), or the percentage of change in $\dot{V}O_2$ after training ($p = 0.460$), indicating a homogeneous population within this group of volunteers. However, ANOVA based on training program

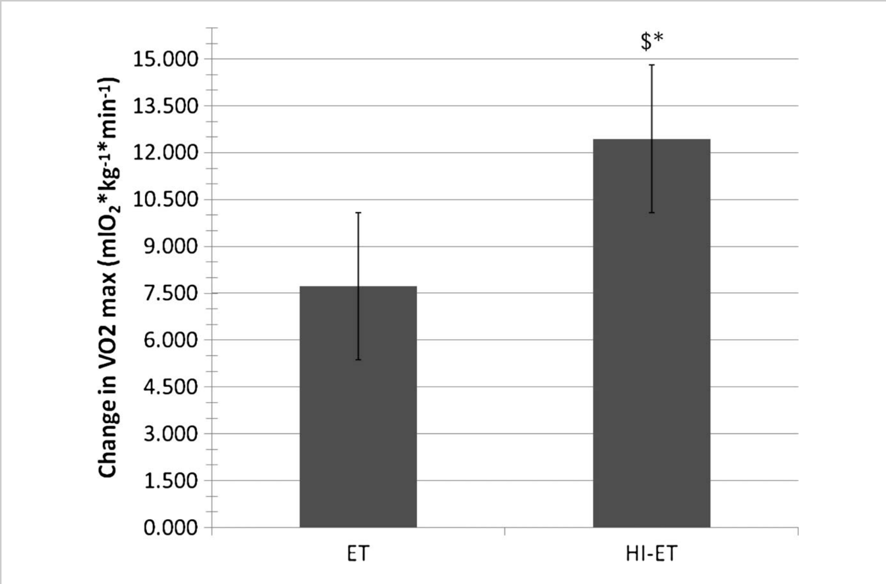

Figure 3. Group average change between pre and posttest measures of estimated $\dot{V}O_2$max for the participants of the 8-week endurance training (ET) program broken into the 2 training methods, ET = continuous-moderate, MI-ET = mixed-intensity interval. *Indicates that the MI-ET group is on average significantly greater than the ET group, and $ indicates that the MI-ET group based on individual change is significantly greater than the ET group after the 8-week training program.

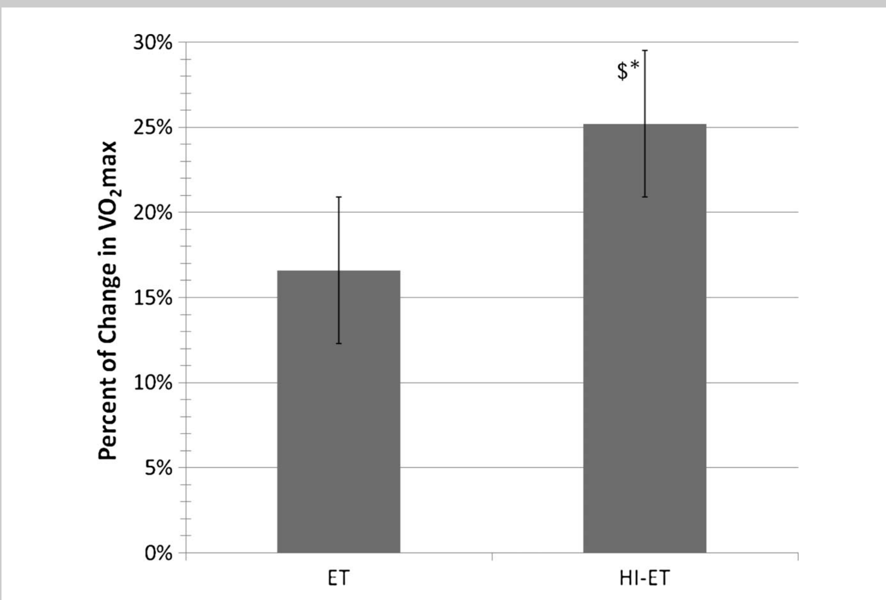

Figure 4. Group average %-change from the pre to posttest measures of estimated $\dot{V}O_2$max for the participants of the 8-week endurance training (ET) program broken into the 2 training methods, ET = continuous-moderate, MI-ET = mixed-intensity interval. *Indicates that the MI-ET group is on average significantly greater than the ET group, and $ indicates that the MI-ET group based on individual change is significantly greater than the ET group after the 8-week training program.

shows that the athletes who participated in the MI-ET program exhibited significantly higher $\dot{V}O_2$ values after training than did the athletes who participated in the traditional ET program ($p = 0.045$).

Although there was limited variance within the whole group, there are differences seen between the groups based on paired *T*-test analysis for group averages for the 2 training paradigms. The paired *T*-test analysis shows that there is further no difference in the pretest $\dot{V}O_2$ ($p = 0.23$) between the MI-ET group of athletes 49.69 ± 1.15 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$ and the ET program, 49.54 ± 1.89 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$. This analysis further indicates that the athletes participating in MI-ET, 62.13 ± 0.96 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$, show significantly higher average posttest $\dot{V}O_2$max (Figure 2) than the ET program, 57.27 ± 1.59 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$ ($p = 0.015$). Furthermore, on average, the MI-ET group exhibited a significantly greater absolute change in estimated $\dot{V}O_2$max 12.44 ± 0.92 ml O_2 $kg^{-1} \cdot min^{-1}$ vs. 7.72 ± 0.99 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$ ($p < 0.001$) (Figure 3). Additionally, the MI-ET group had a significantly greater percentage of change in the $\dot{V}O_2$max ($p < 0.001$) measures $25.2 \pm 2.4\%$ vs. $16.6 \pm 2.6\%$ for the ET group over the same period of time (Figure 4).

Discussion

The physiological demands experienced within a competitive soccer match will force the athlete to vacillate between distinct periods of maximal aerobic capacities efforts with incomplete and intermittent recovery periods between these maximal efforts (1,7). To condition soccer athletes to meet this demand, an interval-training program was designed, the MI-ET program, to mimic these metabolic demands of a competitive soccer match.

The central idea for the MI-ET program was to mimic the demands seen within the soccer game and was accomplished through the development of 3 primary training principles: First, that the training mimics the activities but provides greater physiological demands than seen within the executions of competitive soccer skill (1,5,7,9). Second, that the training provides a demand that encourages positive musculoskeletal and cardiorespiratory responses to increase the aerobic capacity and fitness of the athlete after training (2,4,6,8,10). Finally, that training parameters will provide for carry-over into the athletic competition, the patterning of the exercise effort needs to mirror a competition pattern where submaximal effort would be used before the maximal or near-maximal efforts of the program, mimicking the performance seen within a competitive game (1,7). Additionally, the MI-ET program was carried out on a grass field that nearly replicates a soccer field, with cones randomly dispersed throughout the field to generate a random order of cuts and changes in directions that may be needed within a soccer game (Figure 1).

As would be expected, all volunteers participating in both forms of ET show improvements in the $\dot{V}O_2$ measures after the 8 weeks of training (8). Additionally, given that there is no difference within the group of athletes before training with MI-ET program, improvements in the measurements after training can be attributed to the training program. This difference is seen in the significantly greater posttest $\dot{V}O_2$ measurements after the use of the MI-ET program (62.13 ± 0.96 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$) when compared to the concurrent ET running program (57.27 ± 1.59 ml $O_2 \cdot kg^{-1} \cdot min^{-1}$, $p = 0.015$). This difference between training programs was further observed in the analysis of changes in $\dot{V}O_2$ measures after training, as both an absolute change and the percent of change in $\dot{V}O_2$ after the use of the MI-ET program compared with the group that continued with the concurrent ET program. The changes seen in the volunteer athletes' pre and posttest measures demonstrate that the MI-ET program elicited significantly greater changes and percent increases in $\dot{V}O_2$ levels after training ($p < 0.05$, Figures 3 and 4).

This difference between training regimens is further demonstrated in the comparison of changes elicited here with the findings of previously reported changes using sprint-interval, prolonged-interval, or continuous ET training (4,6,10). In this comparison, the use of the MI-ET program produces greater changes, an average 25.2 ± 2.4% increase, than the previously reported values for changes, an average range of 10–15% increases. At the same time, the changes in $\dot{V}O_2$ for the volunteers that continued the previously established ET program (16.6 ± 2.6% increase after training) show a more favorable comparison ($r = 0.70$) with the same previously reported values (4,6,10).

The differences between the 2 ET programs, after the 8 weeks of training, are most likely attributed to the increased cardiorespiratory demand of the varying the level and intensity of training used within parameters of the MI-ET program. The high intensity, short durations of exertion combined with an incomplete recovery, mimicking the upper level of demands of the soccer match (7), ultimately place an energetic and cardiorespiratory demand on the athletes that required adaptations to occur to meet such high-intensity exercise (6,10). This pattern of exertion and the subsequent adaptations in response to the energetic demands are not seen with the group using the concurrent ET program. This is subjectively observed in the athletes rating the MI-ET exercise demand as a range from 8 to 10 on the Borg Scale of Perceived Exertions, vs. 7 to 8 for the concurrent ET program. This subjective rating is supported by an average HR response of 87.5% of maximal HR during the interval sessions (data not shown) throughout the entire training time of each MI-ET session. This subjective rating and associated HR response are similar to those seen with the short duration intervals that have been traditionally used to improve the cardiorespiratory fitness for various types of athletes (2).

In addition to the higher energetic and cardiorespiratory demand explaining the difference between the ET and the MI-ET programs, some factors may explain the differences that were seen between the MI-ET group and the previously examined groups using a high-intensity ET program (4,6,10). First the MI-ET program employed here was conducted 3 times weekly, whereas the other programs trained only twice weekly and involved longer active intervals that also employed continuous movement (regardless of the level of intensity) for the soccer players throughout the training sessions in this study (4,6,10). Second, because of the desire to allow for testing and retesting by coaches, a 12-minute run test was used to establish aerobic fitness (estimated $\dot{V}O_2$max) while the other studies clinically evaluated a $\dot{V}O_2$max in participants. Even though the 12-minute run test exhibits a high level of reliability and validity to establish an estimated $\dot{V}O_2$max (3), it is not the equivalent of a clinical $\dot{V}O_2$max test that should be used when and where available. These changes in the procedures for testing an individual's $\dot{V}O_2$max may explain a level of error in the differences seen between this group's averages and the group averages of the previously published studies of interval training (4,6,10). While at the same time, the differences in testing procedures should not be dismissed as the sole reason for the differences noted between the group here and with those groups in the previously reported populations.

In conclusion, based on the analysis of the data obtained in the study, the employment of a MI-ET program as designed is more effective than the continuation of the ET for this group of female soccer athletes, or the previous reported sprint-interval and high-intensity interval-training programs at improving the aerobic capacity of a female soccer player. From this vantage point, it can be concluded that the MI-ET program is a valid means to increase one's aerobic capacity.

Practical Applications

In terms of practical applications, the findings from this study suggest that the MI-ET program as designed and executed is a valid and effective means to increase the aerobic fitness of soccer players within a conditioning program. Further, the subjective responses from the athletes suggest that the MI-ET program can easily be implemented into an existing conditioning program, regardless of the mesocycle of training (e.g., off-season, precompetition, or competitive peaking). However tantalizing these results may be, before this conclusion can be generalized to the population as a whole, further research in this MI-ET program is required to examine its usefulness for improvements in general cardiopulmonary fitness in a prolonged study (greater than the 8 weeks observed here). Additionally, it would be of interest to examine changes within a clinical setting to determine the absolute degree to which the MI-ET program can elevate a directly measured $\dot{V}O_2max$, in comparison with the earlier reports for training adaptations.

References

1. Arnason, A, Sigurdsson, SB, Gudmundsson, A, Holme, I, Engebretsen, L, and Bahr, R. Physical fitness, injuries, and team performance in soccer. *Med Sci Sports Exerc* 36: 278–285, 2004.
2. Billat, LV. Interval training for performance: A scientific and empirical practice, special recommendations for middle- and long-distance running. Part I: Aerobic interval training. *Sports Med* 31: 13–31, 2001.
3. Doolittle, TL, Dominic, JA, and Doolittle, J. The reliability of selected cardiorespiratory endurance field test with adolescent female populations. *Am Correct Ther J* 23: 25–33, 1969.
4. Edge, J, Bishop, D, Goodman, C, and Dawson, B. Effects of high- and moderate-intensity training on metabolism and repeated sprints. *Med Sci Sports Exerc* 37: 1975–1982, 2005.
5. Helgerud, J, Engen, LC, Wisloff, U, and Hoff, J. Aerobic endurance training improves soccer performance. *Med Sci Sports Exerc* 33: 1925–1931, 2001.
6. Helgerud, J, Hoydal, K, Wang, E, Karlsen, T, Berg, P, Bjerkaas, M, Simonsen, T, Helgesen, C, Hjorth, N, Bach, R, and Hoff, J. Aerobic high-intensity intervals improve $\dot{V}O_2max$ more than moderate training. *Med Sci Sports Exerc* 39: 665–671, 2007.
7. Hoff, J and Helgerud, J. Endurance and strength training for soccer players, physiological considerations: A review. *Sports Med* 34: 15–180, 2004.
8. Iwasaki, K, Zhang, R, Zuckerman, JH, and Levine, BD. Dose–response relationship of cardiovascular adaptation to endurance training in healthy adults: How much training for what benefit? *J Appl Physiol* 95: 1575–1583, 2003.
9. Little, T and Williams, AG. Suitability of soccer training drills for endurance training. *J Strength Cond Res* 26: 316–319, 2006.
10. Perry, CGR, Heigenhauser, GJF, Bonen, A, and Spriet, LL. High-intensity aerobic interval training increases fat and carbohydrate metabolic capacities in human skeletal muscle. *Appl Physiol Nutr Mtab* 33: 1112–1123, 2008.